

- 单机pg
 - sql1
 - sql

```
select *
from
  gf_tt t1,
  MV_STATION_INFO t2,
  china_county area
where
  t1.v01301=t2.v01301 and t2.netcode='1'
  and t1.d_datetime='2022-03-21 00:00:00'
  and ST_Within(
    t2.point,
    ST_Transform
      (area.geom,4326)
    )=true
  and area.city_name='北京城区';
```

- result

category	index	latency	result num	surf_wea_chn_mul_hor_tab	mv_station_info
local	1	latency average = 80990.667 ms	10056	11689	10143
jdbc_fdw	1	latency average = 89371.180 ms	10056	11689	10143

- sql2

- sql

```
select *
from
  gf_tt t1,
  MV_STATION_INFO t2
where
  t1.v01301=t2.v01301
  and t2.netcode= '1'
  and t1.d_datetime> '2022-01-17'
  and t1.d_datetime <= '2023-01-17 03:00:00'
  and
  ST_DWithin(
    t2.point,
    ST_Transform
      ((select geom from china_river where
        china_river.nam='HAI HE'),4326),5);
```

■ result

category	index	latency	result num	surf_wea_chn_mul_hor_tab	mv_station_info
local	1	latency average = 329.272 ms	10012	11689	10143
jdbc_fdw	1	latency average = 1922.201 ms	10012	11689	10143

• 分布式pg

◦ 环境

- mv_station_info 分布式表 分布式列为v01013
- china_river china_county为复制表

◦ sql1

- sql

```
with t1 as(
    select
        *
    from
        gf_tt
    where
        d_datetime = '2022-03-21 00:00:00'
)
select
    *
from
    t1,
    MV_STATION_INFO t2,
    china_county area
where
    t1.v01301=t2.v01301
    and t2.netcode = '1'
    and ST_Within(
        t2.point,
        ST_Transform
            (area.geom,4326)
    )=true and area.city_name='北京城区';
```

■ result

category	index	latency	result num	surf_wea_chn_mul_hor_tab	mv_station_info
local	1	latency average = 6971.133 ms	10000	11689	10143
jdbc_fdw	1	latency average = 8351.097 ms	10000	11689	10143

■ plan

QUERY PLAN

```
Custom Scan (Citius Adaptive) (cost=0.00..0.00 rows=0 width=0)
(actual time=7161.884..7280.332 rows=10000 loops=1)
  Output: remote_scan.d_retain_id, remote_scan.d_data_id,
remote_scan.d_iymdhm, remote_scan.d_rymdhm,
remote_scan.d_update_time, remote_scan.d_datetime,
remote_scan.v_bbb, remote_scan.v01301, remote_scan.v05001,
remote_scan.v06001, remote_scan.v07001, remote_scan.v0
2175, remote_scan.v07032_03, remote_scan.v08010, remote_scan.v02001,
remote_scan.v02301, remote_scan.v_acode, remote_scan.v04001,
remote_scan.v04002, remote_scan.v04003, remote_scan.v04004,
remote_scan.v04005, remote_scan.v13011, remote_scan.q13011,
remote_scan.d_sou
rce_id, remote_scan.v01301_1, remote_scan.cname,
remote_scan.countrycode, remote_scan.countryname,
remote_scan.regioncode, remote_scan.v_acode_1, remote_scan.v_tcode,
remote_scan.v06001_1, remote_scan.v05001_1, remote_scan.v07001_1,
remote_scan.netcode, remote_scan.v
02301_1, remote_scan.provincename, remote_scan.cityname,
remote_scan.cntyname, remote_scan.townname,
remote_scan.v_acode_4search, remote_scan.d_source_id_1,
remote_scan.point, remote_scan.gid, remote_scan.county_nam,
remote_scan.county_adc, remote_scan.county_cit, re
mote_scan.city_name, remote_scan.city_adcod, remote_scan.city_cityc,
remote_scan.province_n, remote_scan.province_a,
remote_scan.province_c, remote_scan.center, remote_scan.geom
-> Distributed Subplan 2_1
  -> Foreign Scan on public.gf_tt (cost=100.00..111.00
rows=1 width=1022) (actual time=0.110..352.907 rows=11689 loops=1)
    Output: d_retain_id, d_data_id, d_iymdhm, d_rymdhm,
d_update_time, d_datetime, v_bbb, v01301, v05001, v06001, v07001,
v02175, v07032_03, v08010, v02001, v02301, v_acode, v04001, v04002,
v04003, v04004, v04005, v13011, q13011, d_source_id
    Remote SQL: SELECT d_retain_id, d_data_id, d_iymdhm,
d_rymdhm, d_update_time, d_datetime, v_bbb, v01301, v05001, v06001,
v07001, v02175, v07032_03, v08010, v02001, v02301, v_acode, v04001,
v04002, v04003, v04004, v04005, v13011, q13011, d_source_id FRO
```

```

M "SURF_WEA_CHN_MUL_HOR_TAB" WHERE ((d_datetime = '2022-03-21
00:00:00'))
Planning time: 0.000 ms
Execution time: 2086.948 ms
Task Count: 32
Tasks Shown: One of 32
-> Task
      Node: host=172.16.3.123 port=5100 dbname=postgres
      -> Hash Join (cost=925.81..1570.79 rows=5 width=24911)
(actual time=2262.349..2266.477 rows=311 loops=1)
      Output: intermediate_result.d_retain_id,
intermediate_result.d_data_id, intermediate_result.d_iymdhm,
intermediate_result.d_rymdhm, intermediate_result.d_update_time,
intermediate_result.d_datetime, intermediate_result.v_bbb,
intermediate_result.v01301
, intermediate_result.v05001, intermediate_result.v06001,
intermediate_result.v07001, intermediate_result.v02175,
intermediate_result.v07032_03, intermediate_result.v08010,
intermediate_result.v02001, intermediate_result.v02301,
intermediate_result.v_acode, intermedi
ate_result.v04001, intermediate_result.v04002,
intermediate_result.v04003, intermediate_result.v04004,
intermediate_result.v04005, intermediate_result.v13011,
intermediate_result.q13011, intermediate_result.d_source_id,
t2.v01301, t2.cname, t2.countrycode, t2.country
name, t2.regioncode, t2.v_acode, t2.v_tcode, t2.v06001, t2.v05001,
t2.v07001, t2.netcode, t2.v02301, t2.provincename, t2.cityname,
t2.cntyname, t2.townname, t2.v_acode_4search, t2.d_source_id,
t2.point, area.gid, area.county_nam, area.county_adc,
area.county_cit, are
a.city_name, area.city_adcod, area.city_cityc, area.province_n,
area.province_a, area.province_c, area.center, area.geom
      Hash Cond: ((intermediate_result.v01301)::text =
(t2.v01301)::text)
      -> Function Scan on
pg_catalog.read_intermediate_result intermediate_result
(cost=0.00..632.89 rows=3169 width=1022) (actual
time=65.937..66.931 rows=11689 loops=1)
      Output: intermediate_result.d_retain_id,
intermediate_result.d_data_id, intermediate_result.d_iymdhm,
intermediate_result.d_rymdhm, intermediate_result.d_update_time,
intermediate_result.d_datetime, intermediate_result.v_bbb,
intermediate_result.
v01301, intermediate_result.v05001, intermediate_result.v06001,
intermediate_result.v07001, intermediate_result.v02175,
intermediate_result.v07032_03, intermediate_result.v08010,
intermediate_result.v02001, intermediate_result.v02301,
intermediate_result.v_acode, int
ermediate_result.v04001, intermediate_result.v04002,
intermediate_result.v04003, intermediate_result.v04004,
intermediate_result.v04005, intermediate_result.v13011,
intermediate_result.q13011, intermediate_result.d_source_id
      Function Call:
read_intermediate_result('2_1'::text, 'binary'::citus_copy_format)
      -> Hash (cost=925.79..925.79 rows=2 width=23889)
(actual time=2196.384..2196.385 rows=311 loops=1)

```

```

Output: t2.v01301, t2.cname, t2.countrycode,
t2.countryname, t2.regioncode, t2.v_acode, t2.v_tcode, t2.v06001,
t2.v05001, t2.v07001, t2.netcode, t2.v02301, t2.provincename,
t2.cityname, t2.cntyname, t2.townname, t2.v_acode_4search,
t2.d_source_id
, t2.point, area.gid, area.county_nam, area.county_adc,
area.county_cit, area.city_name, area.city_adcod, area.city_cityc,
area.province_n, area.province_a, area.province_c, area.center,
area.geom

Buckets: 1024 Batches: 1 Memory Usage: 69kB
-> Nested Loop (cost=0.00..925.79 rows=2
width=23889) (actual time=36.848..2195.851 rows=311 loops=1)
Output: t2.v01301, t2.cname,
t2.countrycode, t2.countryname, t2.regioncode, t2.v_acode,
t2.v_tcode, t2.v06001, t2.v05001, t2.v07001, t2.netcode, t2.v02301,
t2.provincename, t2.cityname, t2.cntyname, t2.townname,
t2.v_acode_4search, t2.d_sou
rce_id, t2.point, area.gid, area.county_nam, area.county_adc,
area.county_cit, area.city_name, area.city_adcod, area.city_cityc,
area.province_n, area.province_a, area.province_c, area.center,
area.geom

Join Filter: ((st_transform(area.geom,
4326) ~ t2.point) AND _st_contains(st_transform(area.geom, 4326),
t2.point))

Rows Removed by Join Filter: 4713
-> Seq Scan on
public.mv_station_info_102136 t2 (cost=0.00..6.93 rows=314
width=449) (actual time=0.023..0.265 rows=314 loops=1)
Output: t2.v01301, t2.cname,
t2.countrycode, t2.countryname, t2.regioncode, t2.v_acode,
t2.v_tcode, t2.v06001, t2.v05001, t2.v07001, t2.netcode, t2.v02301,
t2.provincename, t2.cityname, t2.cntyname, t2.townname,
t2.v_acode_4search, t2
.d_source_id, t2.point

Filter: ((t2.netcode)::text =
'1'::text)
-> Materialize (cost=0.00..466.74
rows=16 width=23440) (actual time=0.001..0.007 rows=16 loops=314)
Output: area.gid, area.county_nam,
area.county_adc, area.county_cit, area.city_name, area.city_adcod,
area.city_cityc, area.province_n, area.province_a, area.province_c,
area.center, area.geom
-> Seq Scan on
public.china_county_102168 area (cost=0.00..466.66 rows=16
width=23440) (actual time=0.175..1.399 rows=16 loops=1)
Output: area.gid,
area.county_nam, area.county_adc, area.county_cit, area.city_name,
area.city_adcod, area.city_cityc, area.province_n, area.province_a,
area.province_c, area.center, area.geom

Filter:
((area.city_name)::text = '北京城区'::text)

Rows Removed by Filter: 2837

Planning time: 0.732 ms
Execution time: 2267.184 ms

Planning time: 16.293 ms
Execution time: 7377.591 ms
(38 rows)

```

- sql

```
select *
from
    gf_tt t1,
    MV_STATION_INFO t2
where
    t1.v01301=t2.v01301
    and t2.netcode= '1'
    and t1.d_datetime> '2022-01-17'
    and t1.d_datetime <= '2023-01-17 03:00:00'
    and
        ST_DWithin(
            t2.point,
            ST_Transform
                ((select geom from china_river where
china_river.nam='HAI HE'),4326),5);
```

- result

category	index	latency	result num	surf_wea_chn_mul_hor_tab	mv_station_info
local	1	latency average = 613.542 ms	10012	11689	10143
jdbc_fdw	1	latency average = 2635.973 ms	10012	11689	10143

- plan

QUERY PLAN

```
Custom Scan (Citus Adaptive) (cost=0.00..0.00 rows=0 width=0)
(actual time=2193.880..2195.252 rows=10012 loops=1)
```

```

Output: remote_scan.d_retain_id, remote_scan.d_data_id,
remote_scan.d_iymdhm, remote_scan.d_rymdhm,
remote_scan.d_update_time, remote_scan.d_datetime,
remote_scan.v_bbb, remote_scan.v01301, remote_scan.v05001,
remote_scan.v06001, remote_scan.v07001, remote_scan.
v02175, remote_scan.v07032_03, remote_scan.v08010,
remote_scan.v02001, remote_scan.v02301, remote_scan.v_acode,
remote_scan.v04001, remote_scan.v04002, remote_scan.v04003,
remote_scan.v04004, remote_scan.v04005, remote_scan.v13011,
remote_scan.q13011, remote_scan.d
_source_id, remote_scan.v01301_1, remote_scan.cname,
remote_scan.countrycode, remote_scan.countryname,
remote_scan.regioncode, remote_scan.v_acode_1, remote_scan.v_tcode,
remote_scan.v06001_1, remote_scan.v05001_1, remote_scan.v07001_1,
remote_scan.netcode, remote_
scan.v02301_1, remote_scan.provincename, remote_scan.cityname,
remote_scan.cntyname, remote_scan.townname,
remote_scan.v_acode_4search, remote_scan.d_source_id_1,
remote_scan.point
-> Distributed Subplan 1_1
      -> Foreign Scan on public.gf_tt (cost=100.00..111.19
rows=1 width=1022) (actual time=0.114..406.688 rows=11689 loops=1)
          Output: d_retain_id, d_data_id, d_iymdhm, d_rymdhm,
d_update_time, d_datetime, v_bbb, v01301, v05001, v06001, v07001,
v02175, v07032_03, v08010, v02001, v02301, v_acode, v04001, v04002,
v04003, v04004, v04005, v13011, q13011, d_source_id
          Remote SQL: SELECT d_retain_id, d_data_id, d_iymdhm,
d_rymdhm, d_update_time, d_datetime, v_bbb, v01301, v05001, v06001,
v07001, v02175, v07032_03, v08010, v02001, v02301, v_acode, v04001,
v04002, v04003, v04004, v04005, v13011, q13011, d_source_id F
ROM "SURF_WEA_CHN_MUL_HOR_TAB" WHERE ((d_datetime > '2022-01-17
00:00:00')) AND ((d_datetime <= '2023-01-17 03:00:00'))
          Planning time: 0.000 ms
          Execution time: 619.096 ms
          Task Count: 32
          Tasks Shown: One of 32
-> Task
      Node: host=172.16.3.123 port=5100 dbname=postgres
      -> Hash Join (cost=4491.49..5132.81 rows=334 width=1471)
(actual time=98.754..102.397 rows=312 loops=1)
          Output: intermediate_result.d_retain_id,
intermediate_result.d_data_id, intermediate_result.d_iymdhm,
intermediate_result.d_rymdhm, intermediate_result.d_update_time,
intermediate_result.d_datetime, intermediate_result.v_bbb,
intermediate_result.v013
01, intermediate_result.v05001, intermediate_result.v06001,
intermediate_result.v07001, intermediate_result.v02175,
intermediate_result.v07032_03, intermediate_result.v08010,
intermediate_result.v02001, intermediate_result.v02301,
intermediate_result.v_acode, inter
mediate_result.v04001, intermediate_result.v04002,
intermediate_result.v04003, intermediate_result.v04004,
intermediate_result.v04005, intermediate_result.v13011,
intermediate_result.q13011, intermediate_result.d_source_id,
t2.v01301, t2.cname, t2.countrycode, t2.c

```

```

countryname, t2.regioncode, t2.v_acode, t2.v_tcode, t2.v06001,
t2.v05001, t2.v07001, t2.netcode, t2.v02301, t2.provincename,
t2.cityname, t2.cntyname, t2.townname, t2.v_acode_4search,
t2.d_source_id, t2.point
      Inner Unique: true
      Hash Cond: ((intermediate_result.v01301)::text =
(t2.v01301)::text)
      InitPlan 1 (returns $0)
      -> Seq Scan on public.china_river_102169
china_river (cost=0.00..4396.90 rows=9 width=521) (actual
time=11.202..16.965 rows=1 loops=1)
      Output: china_river.geom
      Filter: ((china_river.nam)::text = 'HAI
HE'::text)

      Rows Removed by Filter: 51431
      -> Function Scan on
pg_catalog.read_intermediate_result intermediate_result
      (cost=0.00..632.89 rows=3169 width=1022) (actual
time=68.281..69.300 rows=11689 loops=1)
      Output: intermediate_result.d_retain_id,
intermediate_result.d_data_id, intermediate_result.d_iymdhm,
intermediate_result.d_rymdhm, intermediate_result.d_update_time,
intermediate_result.d_datetime, intermediate_result.v_bbb,
intermediate_resul
t.v01301, intermediate_result.v05001, intermediate_result.v06001,
intermediate_result.v07001, intermediate_result.v02175,
intermediate_result.v07032_03, intermediate_result.v08010,
intermediate_result.v02001, intermediate_result.v02301,
intermediate_result.v_acode,
intermediate_result.v04001, intermediate_result.v04002,
intermediate_result.v04003, intermediate_result.v04004,
intermediate_result.v04005, intermediate_result.v13011,
intermediate_result.q13011, intermediate_result.d_source_id
      Function Call:
read_intermediate_result('1_1'::text, 'binary'::citus_copy_format)
      -> Hash (cost=93.28..93.28 rows=105 width=449)
(actual time=29.823..29.824 rows=312 loops=1)
      Output: t2.v01301, t2.cname, t2.countrycode,
t2.countryname, t2.regioncode, t2.v_acode, t2.v_tcode, t2.v06001,
t2.v05001, t2.v07001, t2.netcode, t2.v02301, t2.provincename,
t2.cityname, t2.cntyname, t2.townname, t2.v_acode_4search,
t2.d_source_
id, t2.point

      Buckets: 1024 Batches: 1 Memory Usage: 32kB
      -> Seq Scan on public.mv_station_info_102136
t2 (cost=0.00..93.28 rows=105 width=449) (actual
time=23.328..29.669 rows=312 loops=1)
      Output: t2.v01301, t2.cname,
t2.countrycode, t2.countryname, t2.regioncode, t2.v_acode,
t2.v_tcode, t2.v06001, t2.v05001, t2.v07001, t2.netcode, t2.v02301,
t2.provincename, t2.cityname, t2.cntyname, t2.townname,
t2.v_acode_4search, t2.d_s
ource_id, t2.point

      Filter: (((t2.netcode)::text = '1'::text)
AND st_dwithin(t2.point, st_transform($0, 4326), '5'::double
precision))

      Rows Removed by Filter: 2
Planning time: 0.594 ms

```


Execution time: 102.969 ms

Planning time: 473.723 ms

Execution time: 2196.739 ms

(35 rows)